$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$$

(i) When $x = 1$

$$2^n = \sum_{k=0}^{n} \binom{n}{k}$$

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n} = 2^n$$

(ii) Differentiate

$$n(1+x)^{n-1} = \sum_{k=1}^{n} k\binom{n}{k} x^{k-1}$$

When $x = 1$

$$n2^{n-1} = \sum_{k=1}^{n} k\binom{n}{k}$$

$$\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n} = n2^{n-1}$$

(iii)

$$\begin{aligned}\int_0^1 (1+x)^n \, dx &= \int_0^1 \sum_{k=0}^{n} \binom{n}{k} x^k \, dx \\ &= \sum_{k=0}^{n} \frac{1}{k+1}\binom{n}{k}\end{aligned}$$

$$\int_0^1 (1+x)^n \, dx = \left[\frac{(1+x)^{n+1}}{n+1}\right]_0^1 = \frac{2^{n+1}-1}{n+1}$$

$$\binom{n}{0} + \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} + \cdots + \frac{1}{n+1}\binom{n}{n} = \frac{2^{n+1}-1}{n+1}$$

(iv) Differentiate twice

$$n(n-1)\big(1+x\big)^{n-2} = \sum_{k=2}^{n} k(k-1)\binom{n}{k}x^{k-2}$$

When $x = 1$

$$\sum_{k=2}^{n} k(k-1)\binom{n}{k} = n(n-1)2^{n-2}$$

$$\sum_{k=1}^{n} k\binom{n}{k} = n2^{n-1}$$

$$\begin{aligned}\sum_{k=1}^{n} k^2\binom{n}{k} &= \sum_{k=2}^{n} k(k-1)\binom{n}{k} - \sum_{k=1}^{n} k\binom{n}{k} \\ &= n(n-1)2^{n-2} + n2^{n-1} \\ &= n(n+1)2^{n-2}\end{aligned}$$

$$\binom{n}{1} + 2^2\binom{n}{2} + 3^2\binom{n}{3} + \cdots + n^2\binom{n}{n} = n(n+1)2^{n-2}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

Last updated: June 25, 2026